

智能仪器设计基础（一）

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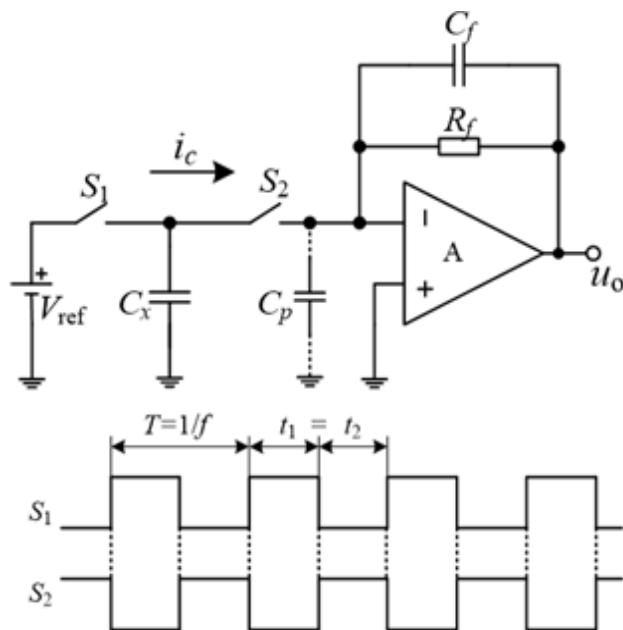
03

智能仪器的前向通道 ——基于理想运放的调理电路设计V

01

基于开关电容法的微小电容的检测（1）

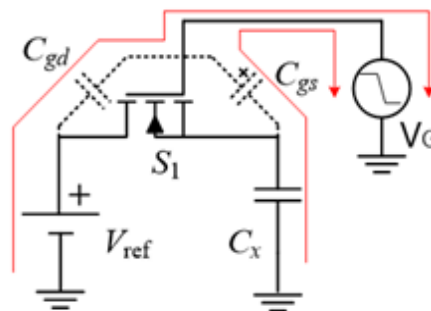
- 微小电容的测量受到杂散电容的影响巨大
- 开关电容技术把电容的测量转换为平均电流的测量
- 运放A消除了寄生电容C_p的影响，但开关存在电荷注入效应



$$\begin{cases} t_1 : S_1 = \text{on}, S_2 = \text{off} \\ \Rightarrow u_x = V_{ref}, Q_x = C_x V_{ref} \\ t_2 : S_1 = \text{off}, S_2 = \text{on} \\ \Rightarrow u_x = 0, Q_x = 0 \end{cases}$$

$$\Rightarrow I_c = \int_0^T i_c dt = \frac{Q_x}{T} = C_x V_{ref} f$$

$$\Rightarrow u_o = -I_c R_f = -R_f C_x V_{ref} f$$

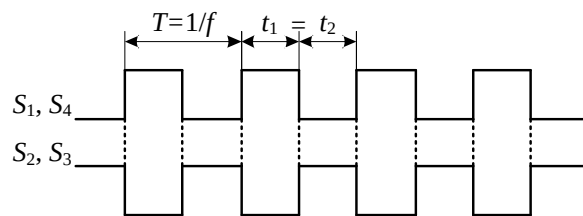
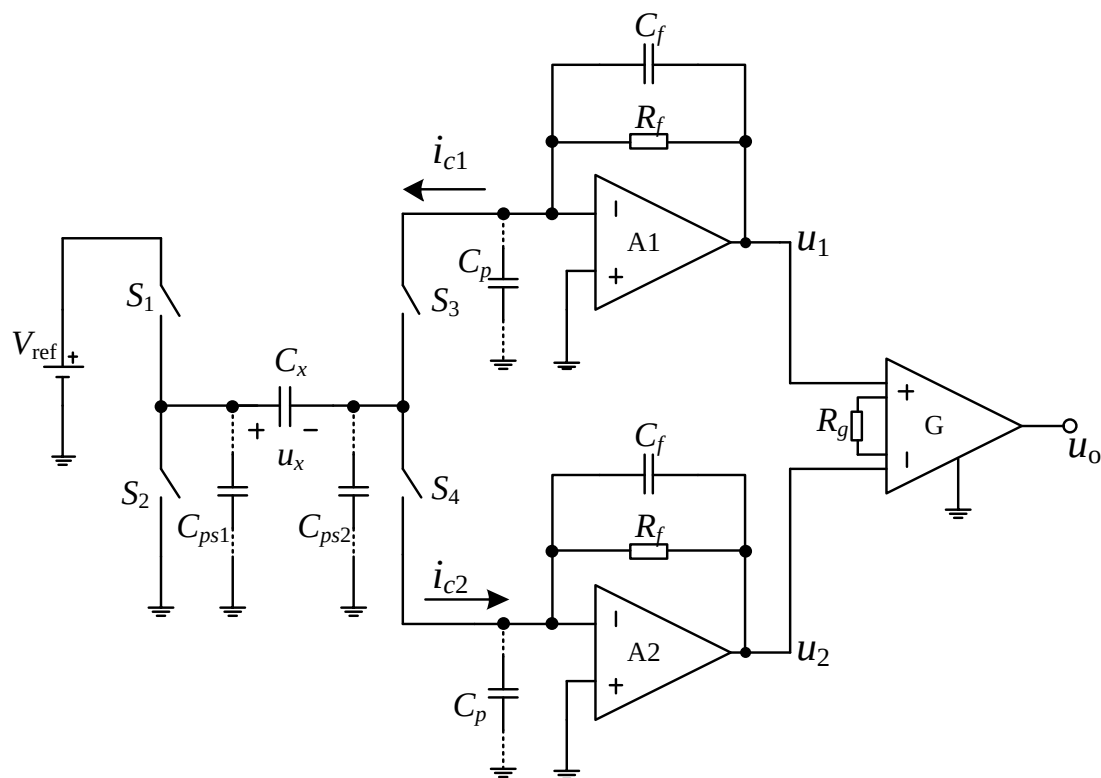


电荷注入效应示意图

01

基于开关电容法的微小电容的检测 (2)

- 差分结构的开关电容电路消除了电荷注入效应的影响
- C_{ps1} 的充放电电流不流经测量电路, C_{ps2} 的影响则被A1和A2消除



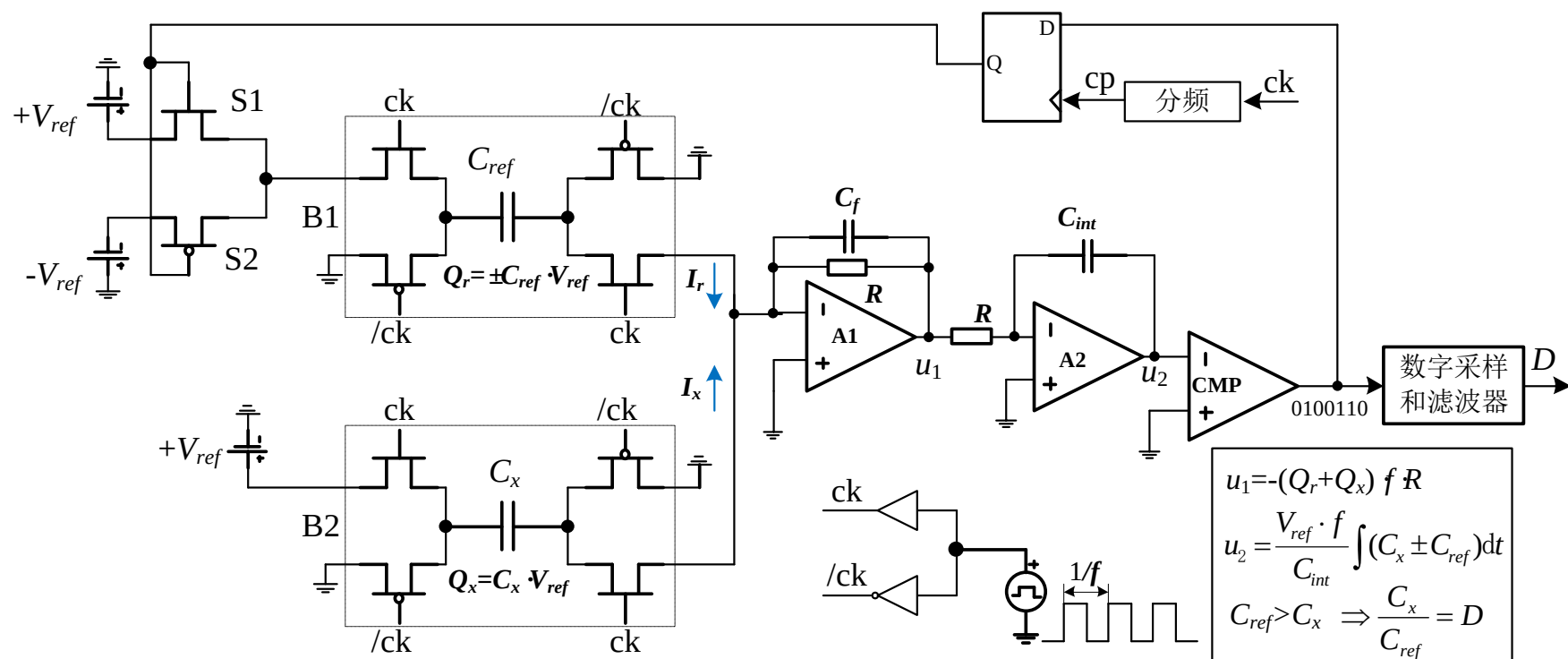
$$\begin{cases}
 t_1 : S_1 = S_4 = \text{on}, S_2 = S_3 = \text{off} \\
 \Rightarrow u_x = 0 \rightarrow V_{ref}, Q_x = C_x V_{ref} \\
 t_2 : S_1 = S_4 = \text{off}, S_2 = S_3 = \text{on} \\
 \Rightarrow u_x = V_{ref} \rightarrow 0, Q_x = -C_x V_{ref}
 \end{cases}$$

$$\Rightarrow I_c = \int_0^T i_c dt = \frac{Q_x}{T} = C_x V_{ref} f$$

$$\Rightarrow u_1 = I_c R_f, u_2 = -I_c R_f$$

$$\Rightarrow u_o = u_1 - u_2 = 2GR_f C_x V_{ref} f$$

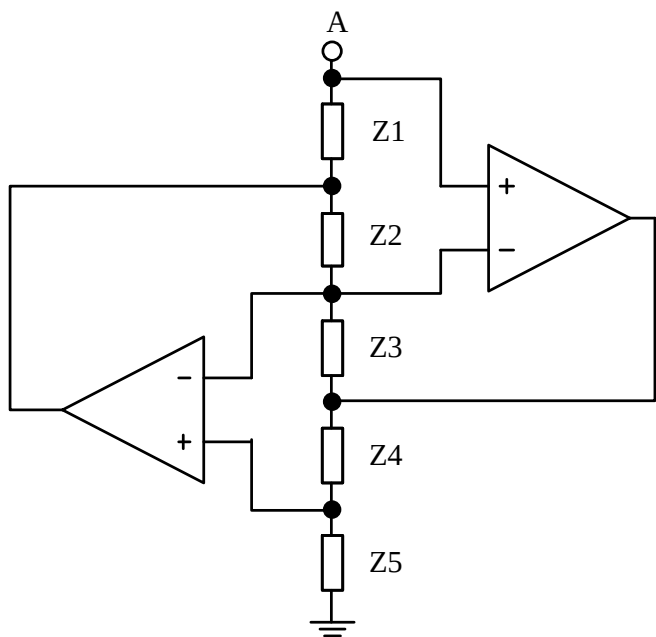
- CDC电容测量将开关电容与 Σ - Δ 模数转换技术相结合实现电容测量和直接数字化



03

通用阻抗产生器—GIC

- 基于运放的有源电路可产生任意阻抗
- GIC通过电阻和电容生成可变电阻、电容、电感以及频变电阻



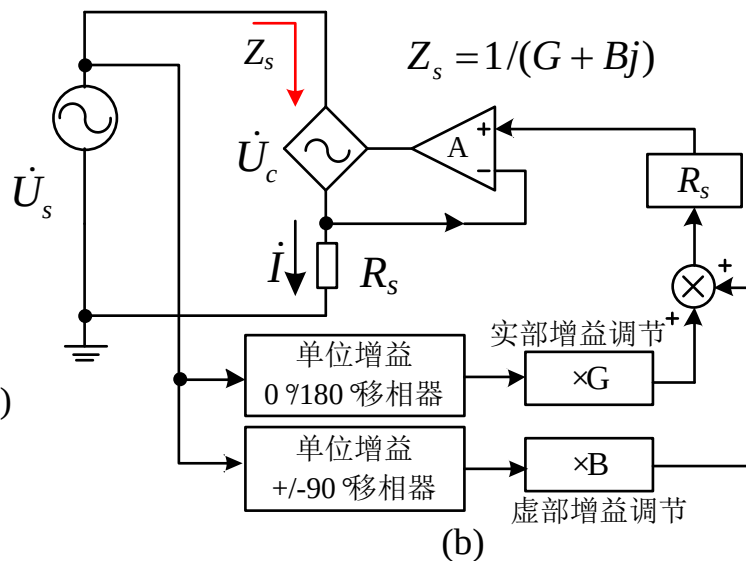
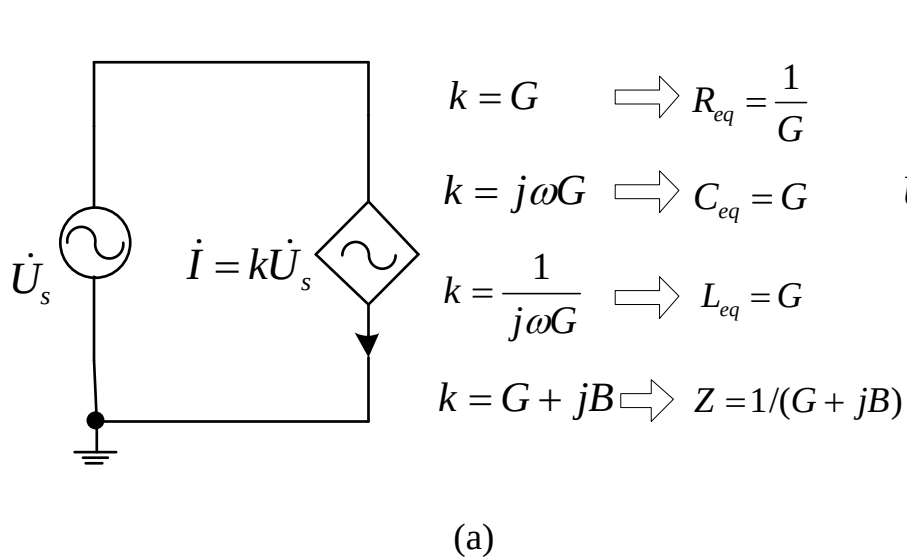
$$Z = \frac{Z_1 Z_3 Z_5}{Z_2 Z_4}$$

- (1) 可调电阻: $Z_x = R_x, Z_5 = R \Rightarrow Z = \frac{R_1 R_3}{R_2 R_4} R = kR = R_{eq}$
- (2) 可调电容: $Z_x = R_x, Z_5 = \frac{1}{j\omega C} \Rightarrow Z = \frac{R_1 R_3}{R_2 R_4} \cdot \frac{1}{j\omega C} = \frac{1}{j\omega kC}, C_{eq} = kC$
- (3) 可调电感: $Z_x = R_x, Z_4 = \frac{1}{j\omega C} \Rightarrow Z = \frac{R_1 R_3 R_5}{R_2} \cdot j\omega C = j\omega kR_3 R_5 C, L_{eq} = kR_3 R_5 C$
- (4) 频变负电阻: $Z_x = R_x, Z_2 = \frac{1}{j\omega C_2}, Z_4 = \frac{1}{j\omega C_4} \Rightarrow Z = -\omega^2 R_1 R_3 R_5 C_2 C_4 = D$

04

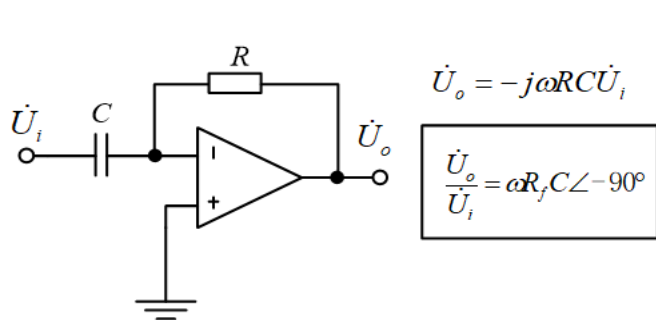
基于受控源的任意阻抗产生电路 (1)

- 交流电源并联受控电流源可构成任意阻抗
- 受控系数为复常数，则阻抗性质决定于实部和虚部
- 实部（与 $\dot{U}_s$ 同相位）表示阻性分量，虚部（ 90° 相差）表示容性和感性分量

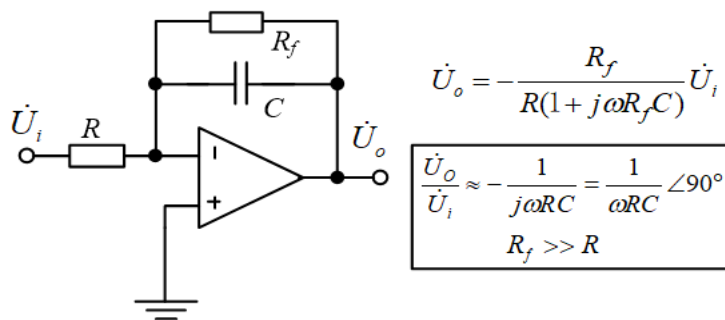


基于受控源的任意阻抗产生电路 (2)

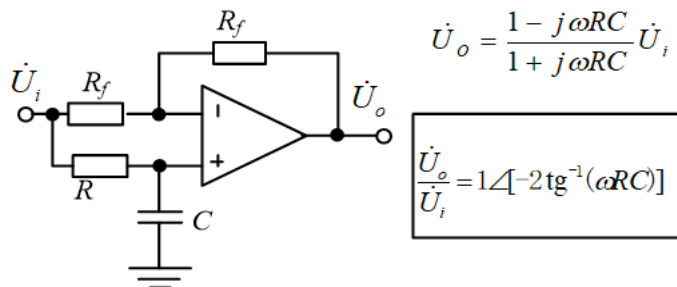
- 0° 和 180° 相移电路决定了阻抗的实部（正电阻或负电阻）
- 90° 相移电路决定了阻抗的虚部（电容或电感）



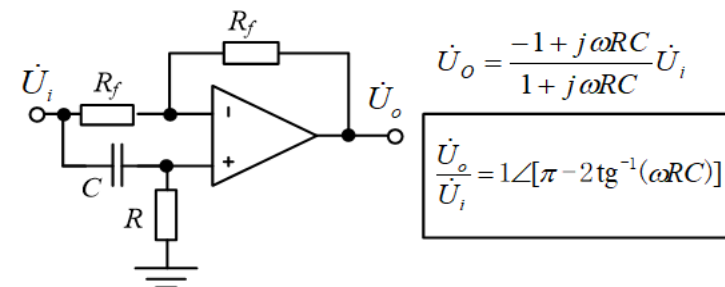
(a)



(b)



(c)



(d)

几种固定频率下的 90° 移相器电路



谢谢观看